

Quasi-Zenith Satellite System
Interface Specification
Positioning Technology Verification Service
(IS-QZSS-TV-004)

(September 27, 2023)

Cabinet Office

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Revision History

Rev. No.	Date	Page	Revisions
001 Draft Edition	March 25, 2016		Draft edition
	July 12, 2016	-	Adds Disclaimer of Liability
001	April 13, 2018	-	First release
002	December 19, 2018	5	• Updates Minimum Signal Strength for QZS-3 • Corrects typographical error(L1S → L5S) in 3.1.8.
003	November 26, 2019	7	Table 3.2-1 • Deletes the explanatory note of PRN Number. • Reassigns PRN Number from 196 to 184. • Reassigns PRN Number from 197 to 189. • Reassigns PRN Number from 200 to 185. • Adds Assignment of PRN Number 186.
		8	Table 3.2-2 • Reassigns PRN Number from 196 to 184. • Reassigns PRN Number from 197 to 189. • Reassigns PRN Number from 200 to 185. • Adds Assignment of PRN Number 186.
004	September 27, 2023	1	• Corrects the number of PS-QZSS versions to 3 in Chapter 1 and 2. • Adds the IS-QZSS-PNT to Applicable Documents.
		2	• Adds description of two operational modes: L5S Normal Mode and L5S Verification Mode to Section 3.1.1. • Adds Section 3.1.1.1. “L5S Normal Mode”.
		3	• Adds description of L5S Verification Mode with Section 3.1.1.2.
		4	• Corrects the label and the number of Occupied Bandwidth in Table 3.1-7. • Corrects to “3.1.3.1. L5S Normal Mode”.
		4-7	• Adds description of “Block III”.
		5	• Adds description of L5S Verification Mode with Section 3.1.3.2.
		6	• Corrects from “Minimum Single Strength” to “Minimum Receiver Power”.
		8	• Corrects to “3.2.1. PRN Number and L5S Mode Assignment” • Adds description of the assignment of operational modes to Section 3.2.1. • Adds “L5S Normal Mode” to the Q5S.
		8	Table 3.2-1 • Adds Assignment of PRN Number 205 and 206. • Adds “L5S Mode” and Remarks corresponding to PRN.

Rev. No.	Date	Page	Revisions
		9	Table 3.2-2 • Adds Assignment of PRN Number 205 and 206.
		10	• Corrects “4.1. L5S Normal Mode”.
		11	• Adds description of CRC Error with Section 4.1.1.5.
		12	• Adds description of “L1C/B” in 4.1.2.2(1)
		14	• Adds description of L5S Verification Mode with Section 4.2.

“TBD” in this document is an abbreviation of “To be determined”. The items marked “TBD” have not been determined yet but will be determined in the future.

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1. Scope

This document describes the interface specifications of the positioning technology verification service (PTV) between the space segment of QZSS and the user segment. The interface specifications described herein include the signal characteristics and message specifications.

The description of system, service and availability is described in the applicable document (1) "PS-QZSS Quasi-Zenith Satellite System Performance Standard".

2. Relevant Documents

2.1. Applicable Documents

The cited parts of the following documents are recognized as being part of this document. This document may be updated when these applicable documents are updated.

- (1) PS-QZSS, Quasi-Zenith Satellite System Performance Standard
- (2) IS-QZSS-PNT, Quasi-Zenith Satellite System Interface Specification Satellite Positioning, Navigation and Timing Service

3. Signal Specifications

3.1. RF Characteristics

3.1.1. Signal Structure

PTV has two operational modes: L5S Normal Mode and L5S Verification Mode. These modes are selected for each satellite transmitting the L5S signal and are operated exclusively because the structure of the L5S signal transmitted in each mode is different.

3.1.1.1. L5S Normal Mode

The signal structure, PRN code characteristic, and message characteristics of L5S Normal Mode are shown in Table 3.1-1, Table 3.1-2, and Table 3.1-3.

Table 3.1-1 Signal Structure

Frequency Band	Signal Name	Modulation Method	PRN Code Name	Overlay Code Name	Message Name
L5	L5S	QPSK	I5S	-	L5S
			Q5S	Neuman-Hoffman	-

Table 3.1-2 PRN Code Characteristics

PRN Code Name	Chip Rate	Length	Period	Overlay Code
I5S	10.23 Mcps	10,230 chips	1 ms	-
Q5S	10.23 Mcps	10,230 chips	1 ms	Neuman-Hoffman Length:20bits Period:1ms

Table 3.1-3 Message Characteristics

Message Name	Bit Rate	Symbol Rate	Period (Minimum Frame)	Encoding Method
L5S	250 bps	500 sps	1 sec	CRC Convolutional code

3.1.1.2. L5S Verification Mode

The signal structure, PRN code characteristic, and message characteristics of L5S Verification Mode are shown in Table 3.1-4, Table 3.1-5, and Table 3.1-6.

Table 3.1-4 Signal Structure

Frequency Band	Signal Name	Modulation Method	PRN Code Name	Overlay Code Name	Message Name
L5	L5S	QPSK	I5S	-	L5S(I-ch)
			Q5S	-	L5S(Q-ch)

Table 3.1-5 PRN Code Characteristics

PRN Code Name	Chip Rate	Length	Period	Overlay Code
I5S	10.23 Mcps	10,230 chips	1 ms	-
Q5S	10.23 Mcps	10,230 chips	1 ms	-

Table 3.1-6 Message Characteristics

Message Name	Bit Rate	Symbol Rate	Period (Minimum Frame)	Encoding Method
L5S	250 bps	1,000 sps	1 sec	CRC Convolutional code Manchester coding

3.1.2. Frequency

The frequency band, nominal center frequency and occupied bandwidth of L5S signals are shown in Table 3.1-7.

However, the reference frequency (f_0) = 10.23 MHz is offset by the nominal $\Delta f/f_0 = -5.399\text{E-}10$ to compensate for the frequency difference between the ground surface and satellite orbit due to the relativistic effect. For this reason, the center frequency in the satellite orbit is not exactly precise. For example, the L5 band signal is offset by -0.6352Hz (nominal).

Table 3.1-7 Center Frequency and Occupied Bandwidth

Frequency Band (Signal Name)	Nominal center frequency	Occupied Bandwidth
L5 band (L5S)	1176.45 MHz	24.0 MHz (± 12.00 MHz)

3.1.3. Modulation Method

3.1.3.1. L5S Normal Mode

L5S (Normal Mode) signals are modulated by QPSK. The modulation method is shown in Figure 3.1-1.

L5S navigation messages and PRN codes are modulated by exclusive-OR (modulo 2 addition) and then modulated with L5 carrier waves by QPSK.

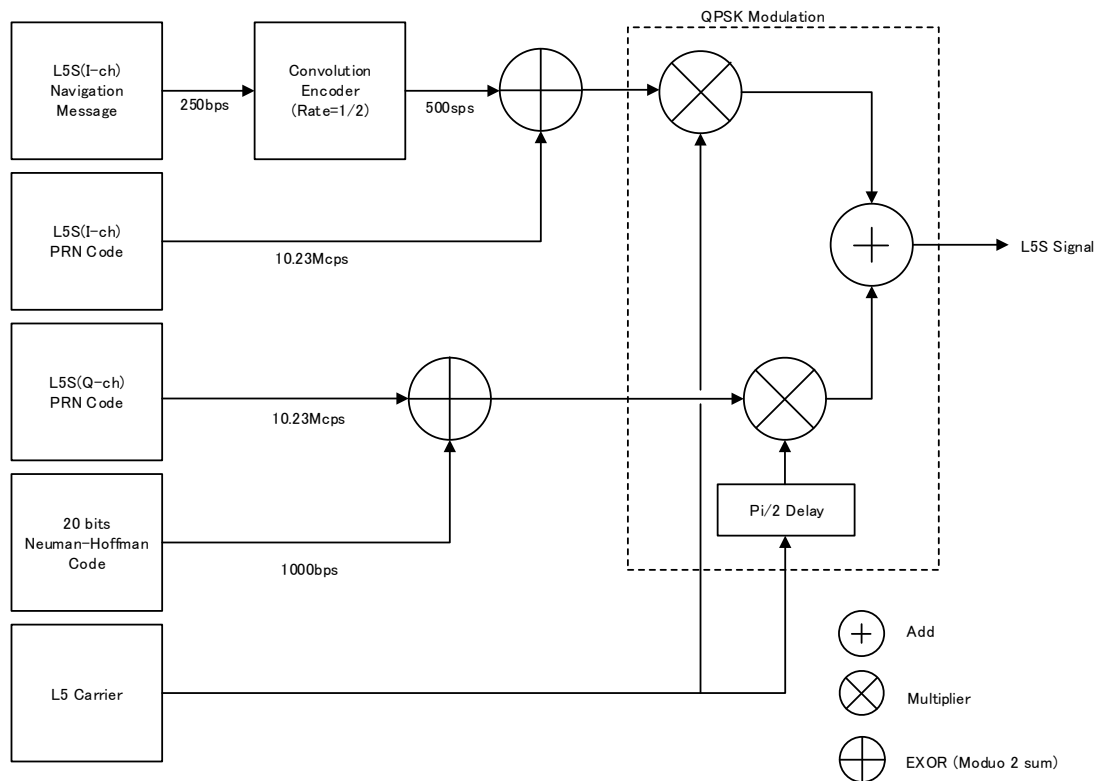


Figure 3.1-1 L5S (Normal Mode) Modulation

3.1.3.2. L5S Verification Mode

L5S (Verification Mode) signals are modulated by QPSK. The modulation method is shown in Figure 3.1-2.

L5S navigation messages and PRN codes are modulated by exclusive-OR (modulo 2 addition) and then modulated with L5 carrier waves by QPSK.

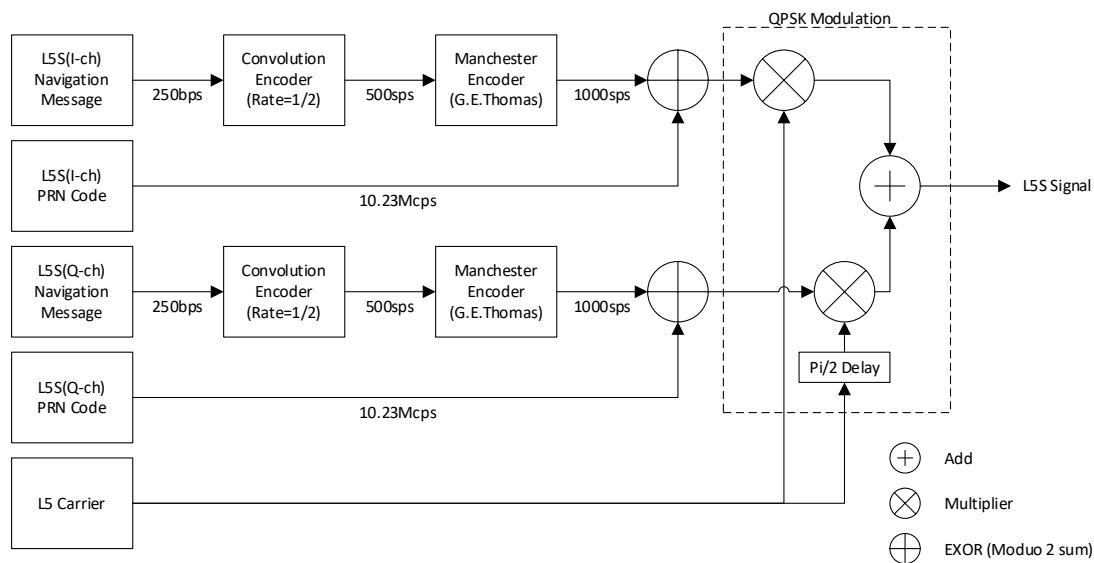


Figure 3.1-2 L5S (Verification Mode) Modulation

3.1.4. Correlation Loss

For L5S signals, correlation loss is defined as the energy ratio of carrier wave to output signals obtained by theoretically modulating the output signals by reverse diffusion.

The correlation loss values are as shown below:

- Block II and III: 1.0 dB or less

3.1.5. Carrier Phase Noise

For L5S signals, carrier phase noise is defined as the spectral density of the phase noise of the unmodulated carrier wave before the superposition of the PRN code and navigation message.

The carrier phase noise shall be such that a phase-locked loop (PLL) with a 10Hz single-sided bandwidth will be able to track the carrier phase to the following values:

- Block II and III: 0.1 rad (RMS)

3.1.6. Spurious

For L5S signals, spurious transmission is defined as unwanted signals in the unmodulated carrier wave before the superposition of the PRN code and navigation message.

The spurious transmission shall be as follows:

- Block II and III: -40 dB or less

3.1.7. Phase Relationship within Signals

For L5S signals, the phase relationships between I-channel and Q-channel of L5S are shown in Figure 3.1-3.

The I-channel had 90° phase lag from the Q-channel. The accuracy is $\pm 5^\circ$.

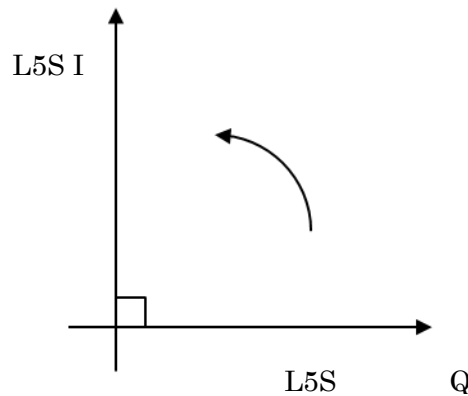


Figure 3.1-3 Phase relationship of L5S

3.1.8. Minimum Received Power

The minimum received power is measured at a ground-based isotropic antenna with a gain of 0dBi for circularly polarized wave reception, when L5S signals are received from a satellite with an elevation angle of 10° or more. The power is shown in Table 3.1-8.

Table 3.1-8 Minimum Received Power

Signal Name	Block II and III
L5S	SV Number =3 : -162.6dBW Other SV Number : -157.0 dBW

3.1.9. Polarization Property

L5S signals are right-hand circularly polarized.

At the center frequency of each signal, the axial ratio (power ratio of the long axis to short axis) of the ellipse of the circularly polarized wave is within the beam range $\pm 10^\circ$ from the boresight direction. The axial ratio is as shown below:

- Block II and III: 2.0 dB or less

3.1.10. Group Delay Property

3.1.10.1. Group Delay Between Signals

L5S signals have, no group delay between signals.

3.1.10.2. Group Delay Between Signals of the Same Frequency

L5S signals have no group delay between signals of same frequency.

For L5S signals, no specifications are defined for the group delay between signals of the same frequencies.

3.1.11. PRN Code Jitter

The jitter with the PRN code zero-crossing interval shall be as follows:

- 2.0 ns or less (3σ).

For PRN codes, the average time difference between the rising edge and the falling edge shall be as follows:

- 1.0 ns or less

3.1.12. Code Carrier Coherence

(1) Short-term code carrier coherence

The short-term (less than 10 sec) difference between the code pseudo range rate and the Doppler frequency shall be as follows:

- 0.015 m/sec or less (1σ).

(2) Long-term code carrier coherence

The long-term (less than 100 sec) difference between the code phase pseudo range and the carrier phase pseudo range shall be as follows:

- 0.255 m or less (1σ).

(3) Short-term L1/L5 coherence

The short-term (less than 10 sec) difference between the L1 Doppler frequency and the L5 Doppler frequency shall be as follows:

- 0.015 m/sec or less (1σ).

(4) Long-term L1/L5 coherence

The long-term (less than 100 sec) difference between the L1 carrier phase pseudo range and the L5 carrier phase pseudo range shall be as follows:

- 0.255 m or less (1σ).

3.2.1. PRN Number and L5S Mode Assignment

Table 3.2-1 Assignment of PRN Number and L5S Mode by satellite categories

PRN Number	Satellite Category	Block Assignment	SV Number	L5S Mode	Remarks
184	QZO	Block IIQ	2	Normal	QZS02
189	GEO	Block IIG	3	Normal	QZS03
185	QZO	Block IIQ	4	Normal	QZS04
186	QZO	Block IIA-Q	5	Normal or Verification (*)	QZS1R
205	GEO	Block III	7	Verification	QZS06
206	QGEO	Block III	8	Verification	QZS07

3.2.2. L5S Code

1st	Last
20-bit NH code = 0000 0100 1101 0100 1110	

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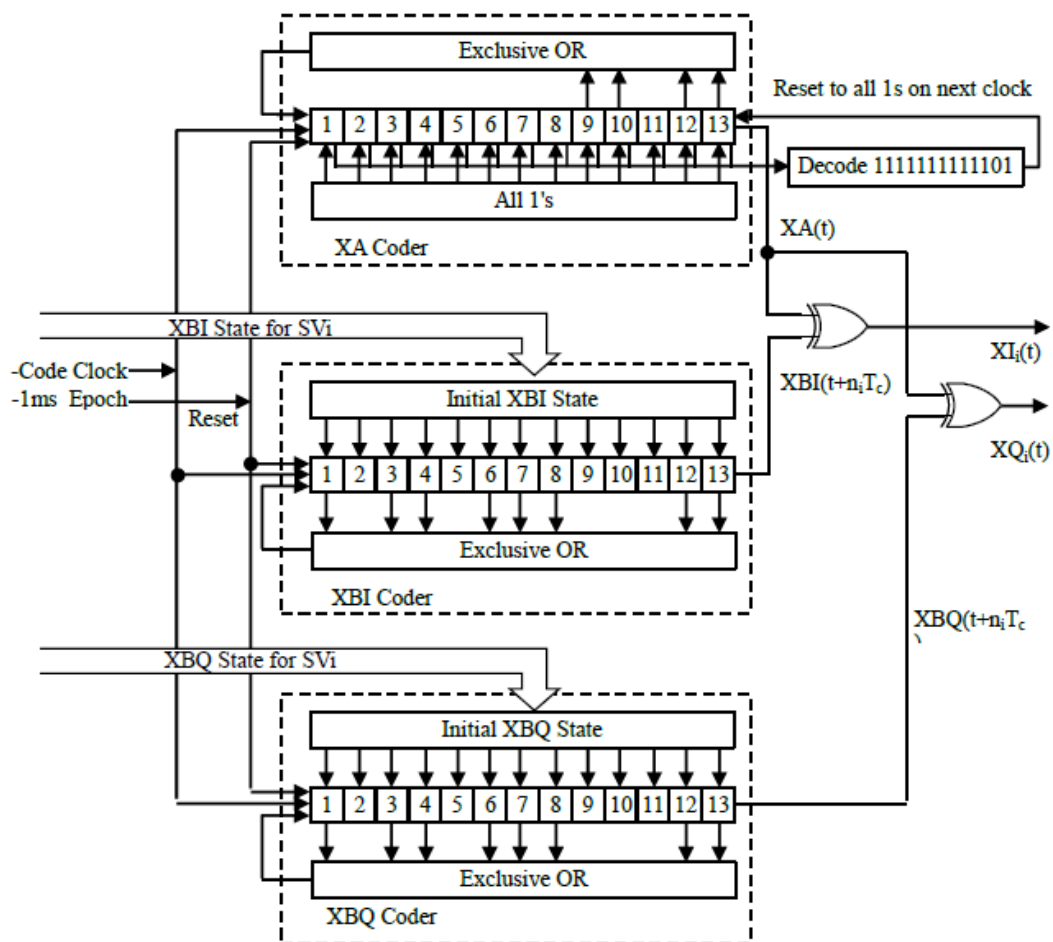


Figure 3.2-1 L5S Code Generator

Table 3.2-2 L5S PRN Code phase assignment

PRN Number	XB Code Advance (Chips)		Initial XB Code State	
	I5	Q5	I5	Q5
184	0645	0977	1111000100010	0011111111000
189	0923	1266	0010011111101	1101011100011
185	2401	0681	0101001000011	0010001101101
186	1639	4273	0011101111100	0001100110011
205	6948	0037	0010111000001	1000001110100
206	4664	1943	1101100001111	1000010101100

3.2.3. Non-Standard Codes

L5S signals have no non-standard codes.

4. Message Specifications

4.1. L5S Normal Mode

4.1.1. Message Configuration

4.1.1.1. Overview

Each message is transmitted by L5S signal consists of 250 bits shown in Figure 4.1-1. The single message is transmitted in one second (250bps).

Each message consists of 8-bit preamble (PAB),6-bit message type (MT), 212-bit data field (DATA FIELD) and a 24-bit CRC (CRC). Since the message transmission sequence is not specified, any message may be transmitted in each one-second period.

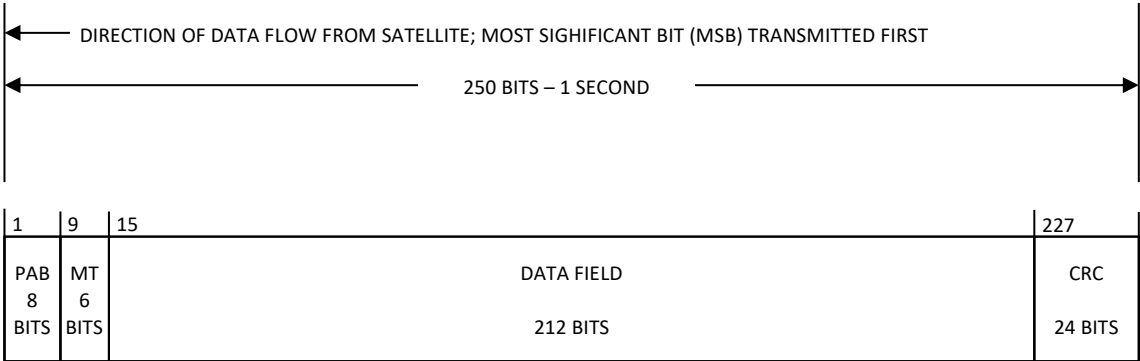


Figure 4.1-1 Message Block Format

4.1.1.2. Timing

(1) Transmission Satellite

PTV message may be different for each satellite.

(2) Transmission Interval

The transmission interval depends on the number of PTV users.

4.1.1.3. Cyclic Redundancy Check (CRC)

The 24-bit CRC bit string is generated by the following generator polynomial $g(X)$

$$g(X) = X^{24} + X^{23} + X^{18} + X^{17} + X^{14} + X^{11} + X^{10} + X^7 + X^6 + X^5 + X^4 + X^3 + X + 1$$

4.1.1.4. Forward Error Correction (FEC)

The bit string will be Forward Error Correction (FEC) encoded by a 1/2 rate convolutional code. 250-bps messages are encoded to 500-sps symbols. The convolutional coding will have a constraint length of 7, with a convolutional encoder logic arrangement as shown in Figure 4.1-2. The G1 symbol is selected on the output as the first half of a 4-millisecond data bit period and the G2 symbol is the second half.

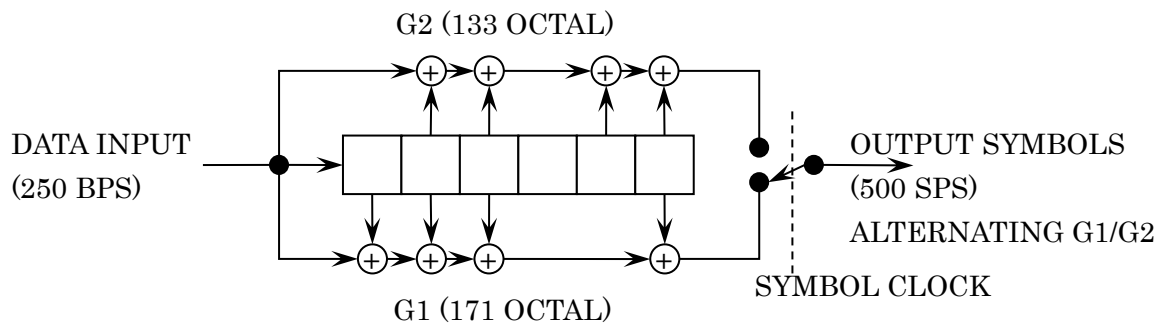


Figure 4.1-2 FEC method

4.1.1.5. CRC Error Message

If there is no information to be transmitted, as determined by the satellite system side, the message with CRC error is transmitted intentionally.

4.1.2. Message Contents

4.1.2.1. Overview

As described in Section 4.1.1.1. , each PTV message provided by the L5S signal consists of a preamble, a message type, a data field, and a CRC parity. PTV message field shall be defined by the users of this service.

4.1.2.2. Common Section

All message types contain the common parameters as shown in Figure 4.1-3 and Table 4.1-1.

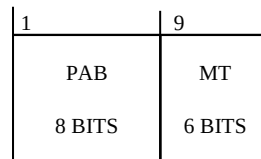


Figure 4.1-3 Common Sections Format

Table 4.1-1 Parameter Definitions of Common Sections

Parameter	Description	Effective Range	Number of Bits	LSB	Units
PAB	Preamble	-	8	-	-
MT	Message type	0-63	6	1	-

(1) Preamble

The beginning of each message is the 8-bit preamble that consists of the following three patterns repeated in sequence:

Pattern A: 01010011
Pattern B: 10011010
Pattern C: 11000110

The first bit in the "Pattern A" preamble is synchronous with the epoch of the 6-second L1C/A or L1C/B signal (see applicable document (2) "IS-QZSS-PNT") navigation message subframe. "Pattern B" comes after "Pattern A". "Pattern C" comes after "Pattern B". After that, the sequence returns to "Pattern A".

FEC encoding is transformed for preambles in the same coding (See Section 4.1.1.4.). While the preamble indicates the beginning of the message block, it cannot be used for signal acquisition or for bit synchronization prior to FEC decoding.

(2) Message type

Message type shall be defined by PTV users. However, Message type 63 shall be used as

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a null message.

4.1.2.3. Message Type 63: Null Message

Figure 4.1-4 shows the data format and Table 4.1-2 shows its parameter definitions.

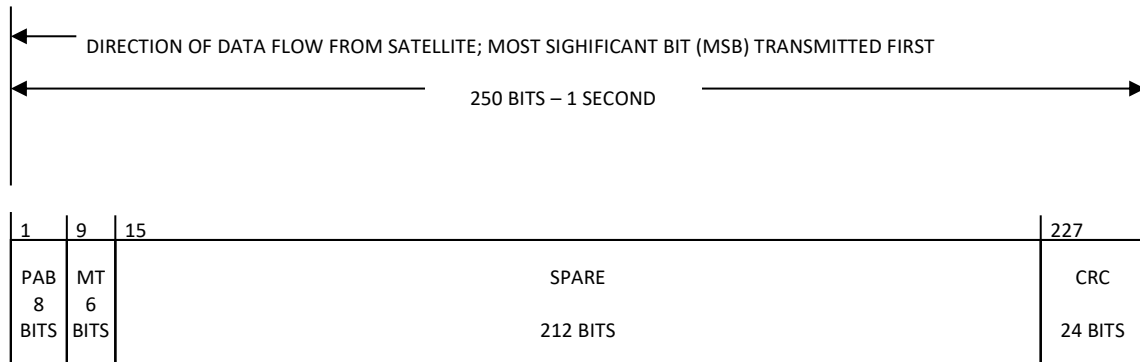


Figure 4.1-4 Message Type 63: Null Message Format

Table 4.1-2 Parameter Definitions

Parameter	Description	Effective Range	Number of Bits	LSB	Units
-	PAB and MT (see 4.1.2.2.)	-	-	-	-
Spare	Spare	-	212	1	-
-	CRC (see 4.1.1.3.)	-	-	-	-

(1) Spare

Fix to "0" for spare.

4.2. L5S Verification Mode

4.2.1. Message Configuration

4.2.1.1. Overview

Each message is transmitted by L5S signal consists of 250 bits shown in Figure 4.2-1. The single message is transmitted in one second (250bps).

Each message consists of 4-bit preamble (PAB),6-bit message type (MT), 216-bit data field (DATA FIELD) and a 24-bit CRC (CRC). The message transmission sequence is not specified, so any message may be transmitted in each one-second period.

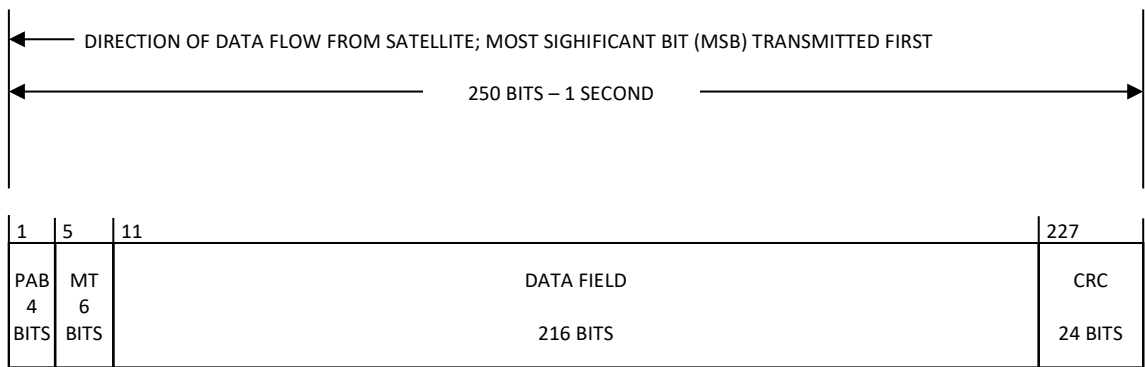


Figure 4.2-1 Message Block Format

4.2.1.2. Timing

(1) Transmission Satellite

See 4.1.1.2. (1) .

(2) Transmission Interval

See 4.1.1.2. (2) .

4.2.1.3. Cyclic Redundancy Check (CRC)

See 4.1.1.3.

4.2.1.4. Forward Error Correction (FEC)

See 4.1.1.4.

4.2.1.5. Manchester Coding

The L5S signals use Manchester encoding as per G.E. Thomas. Figure 4.2-2 shows the convention of the Manchester encoding, where a “0” is expressed by a low-to-high transition (“0” during the first half of the bit period and “1” during the second half) and a “1” is expressed by a high-to-low transition (“1” during the first half of the bit period and “0” during the second half).

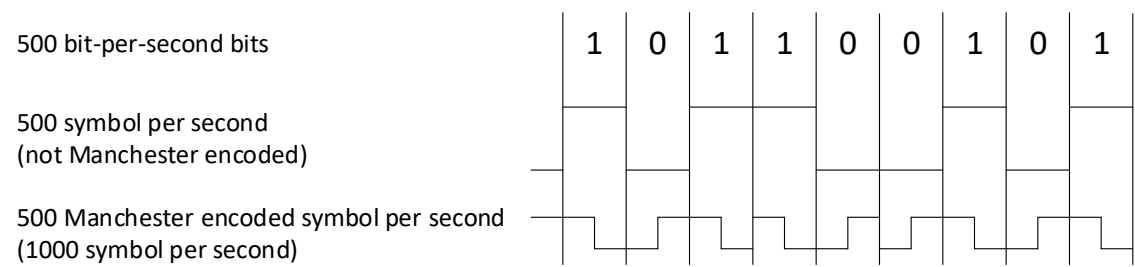


Figure 4.2-2 Manchester Coding method

4.2.1.6. Null Message and CRC Error Message

If there is no information to be transmitted, as determined by the satellite system side, Null Message (See 4.2.2.3.) is transmitted by the QZS with SV Number 5, and the message with CRC error is transmitted intentionally by the QZSs with SV Number 7 and 8.

4.2.2. Message Contents

4.2.2.1. Overview

As described in Section 4.2.1.1. , each PTV message provided by the L5S signal consists of a preamble, a message type, a data field, and a CRC parity. PTV message field shall be defined by the users of this service.

4.2.2.2. Common Section

All message types contain the common parameters as shown in Figure 4.2-3 and Table 4.2-1.

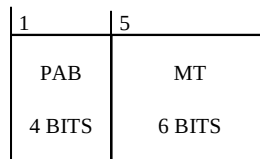


Figure 4.2-3 Common Sections Format

Table 4.2-1 Parameter Definitions of Common Sections

Parameter	Description	Effective Range	Number of Bits	LSB	Units
PAB	Preamble	-	4	-	-
MT	Message type	0-63	6	1	-

(1) Preamble

The beginning of each message is the 4-bit preamble that consists of the following three patterns repeated in sequence:

Pattern A:	0101
Pattern B:	1100
Pattern C:	0110
Pattern D:	1001
Pattern E:	0011
Pattern F:	1010

The first bit in the "Pattern A" preamble is synchronous with the epoch of the 6-second L1C/A or L1C/B signal (the applicable document (2) "IS-QZSS-PNT") navigation message subframe. "Pattern B" comes after "Pattern A". "Pattern C" comes after "Pattern B". After that, the sequence goes through "Pattern D", "Pattern E", and "Pattern F" and back to "Pattern A".

FEC encoding is transformed for preambles in the same coding (See Section 4.2.1.4.). Accordingly, while the preamble indicates the beginning of the message block, it cannot be used for signal acquisition prior to FEC decoding or for bit synchronization.

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(2) Message type

As in Chapter 4.1.2.2. (2) , Message type shall be defined by PTV users. However, Message type 63 shall be used as a null message.

4.2.2.3. Message Type 63: Null Message

Figure 4.2-4 shows the data format and Table 4.1-2 shows its parameter definitions.

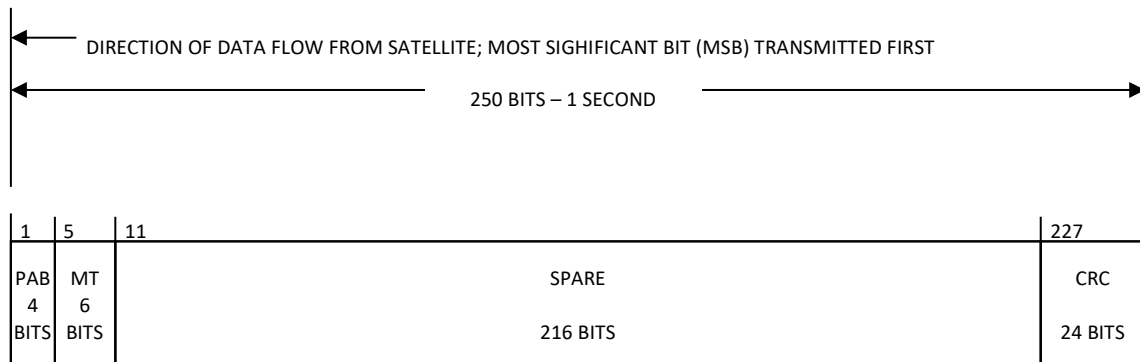


Figure 4.2-4 Message Type 63: Null Message Format

Table 4.2-2 Parameter Definitions

Parameter	Description	Effective Range	Number of Bits	LSB	Units
-	PAB and MT (see 4.2.2.2.)	-	-	-	-
Spare	Spare	-	216	1	-
-	CRC (see 4.2.1.3.)	-	-	-	-

(1) Spare

Fix to "0" for spare.